Failure Analysis Report

FAIL-0285 — Crash

1. Failure Summary

Failure ID	FAIL-0285
Failure Type	Crash
Severity	Minor
Component	Notification Handler
Application	APP-010 — MediaPlatform
Build Version	4.5.9
Timestamp	2024-11-25 15:39:53
Environment	ENV-009 — Low Network Env
OS Version	Android 14
Network	Low-BW Sim
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-0281
Title	Test that NFC tap-to-pay correctly handles invalid input when the input data con...
Execution ID	EXEC-01362
Test Run	RUN-002 — Regression Cycle 2 — Build 4.2.1
Requirement	REQ-0295 — Network Behavior — Offline Mode

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete voice command recognition within the defined SLA.
Observed Behaviour	The system exhibited crash when the input data contains Unicode special characters.

4. Error Details

Error Message:

```
Process com.samsung.wallet terminated unexpectedly with signal SIGSEGV (segmentation fault).
```

Stack Trace / Log Excerpt:

```
com.samsung.health.Diagnostics.execute(line 150)
```

Log Reference	LOG-79227

5. Executed Steps at Time of Failure

1	Launch the SmartHome application on the test device.	The performance counter logs a value below the SLA threshold.

SYNTHETIC DATA — No real Samsung confidential information

2	Initiate payment processing by pressing the 'Connect' button.	The fallback UI is displayed with a human-readable error message.
3	Disconnect the phone and wait 27 seconds.	The operation completes without error.
4	Disconnect the headset and wait 8 seconds.	A success toast message is displayed.

6. Root Cause Analysis

Firmware update binary not validated against checksum before flashing; partial write leaves device in boot loop.

8. Historical Similar Failures

HIST-0124	FAIL-0125	Crash	5.1.8	2022-03-30	8	Resolved
HIST-0284	FAIL-0285	Crash	5.4.6	2021-11-20	13	Resolved
HIST-1224	FAIL-0125	Crash	5.1.5	2022-05-23	14	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information